EDS Practical – 3

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3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, and , representing the number of rows and the number of columns.
- The next lines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

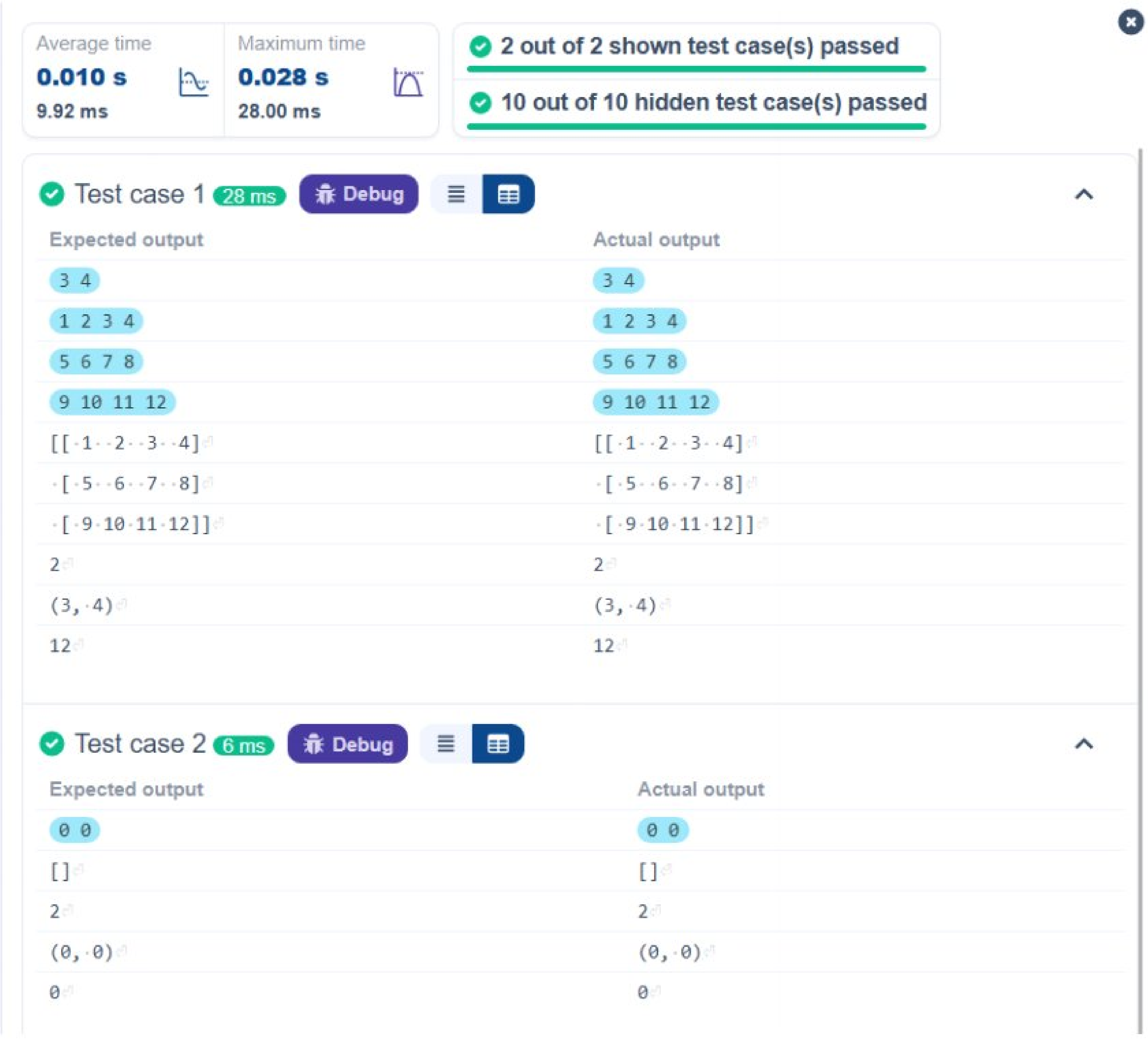
Note: Use reshape() function to reshape the input array with the specified number of rows and columns.

Code:

```
import numpy as np

r, c = map(int, input().split()

elements = []

for _ in range(r):
```

```
        elements.extend(list(map(int, input().split())))
arr = np.array(elements).reshape(r, c)
print(arr)
print(arr.ndim)
print(arr.shape)
print(arr.size)
```

Output:



Lab Assignment

3.2.1. Numpy: Matrix Operations

The given code takes two 3X3 matrices, matrix a and matrix b as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** ()
2. **Subtraction** ()
3. **Element-wise Multiplication** ()
4. **Matrix Multiplication** ()
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

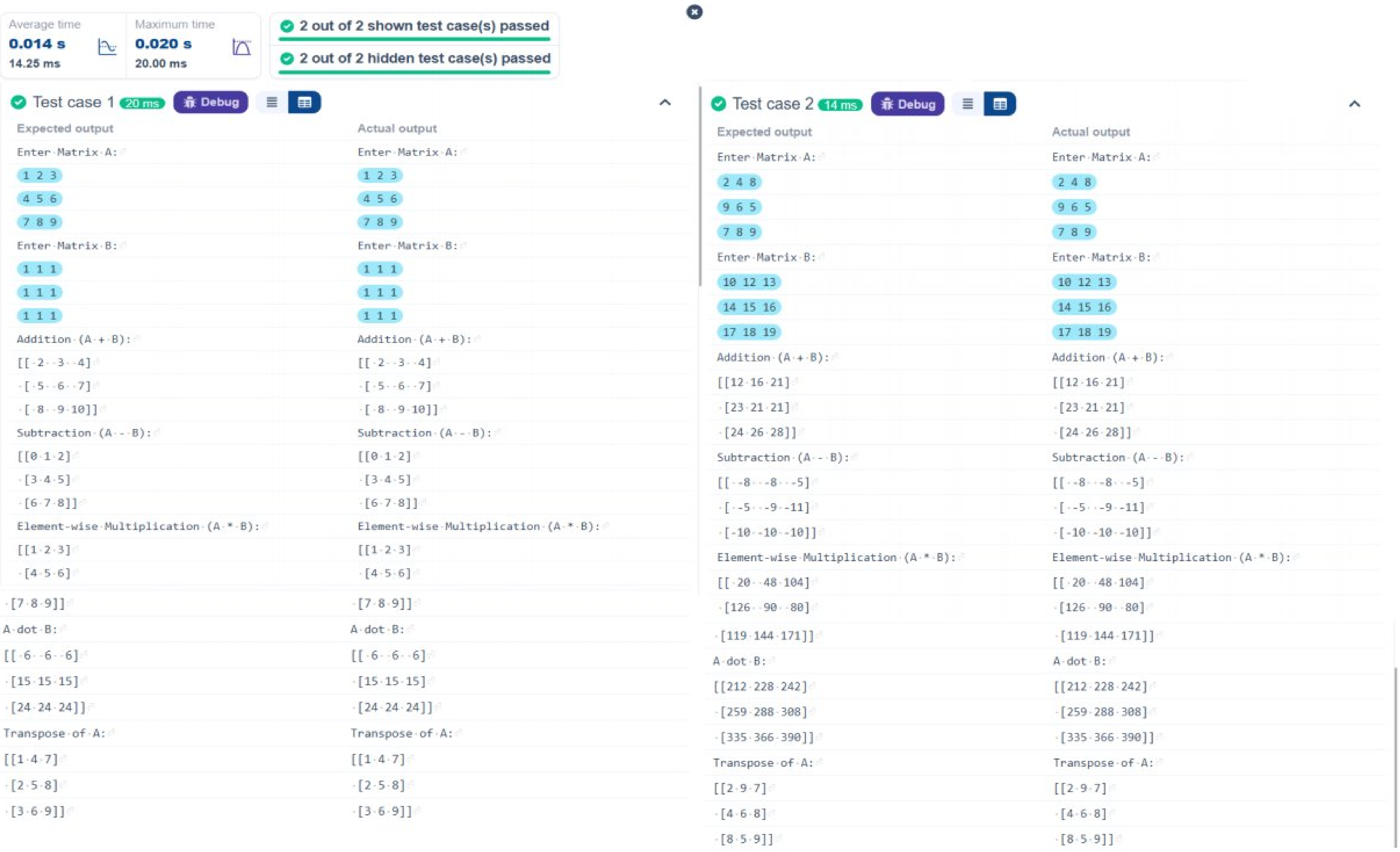
The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A

Code:

```
import numpy as np

# Input matrices

print("Enter Matrix A:")

matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")

matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

# Addition

addition_result = matrix_a + matrix_b

print("Addition (A + B):")

print(addition_result)

# Subtraction

subtraction_result = matrix_a - matrix_b

print("Subtraction (A - B):")

print(subtraction_result)

# Multiplication (element-wise)

elementwise_multiplication_result = matrix_a * matrix_b

print("Element-wise Multiplication (A * B):")

print(elementwise_multiplication_result)

# Matrix multiplication (dot product)

matrix_multiplication_result=np.dot(matrix_a,matrix_b)

print("A dot B:")

print(matrix_multiplication_result)

# Transpose

transpose_a=matrix_a.T

print("Transpose of A:")

print(transpose_a)
```

Output:



3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

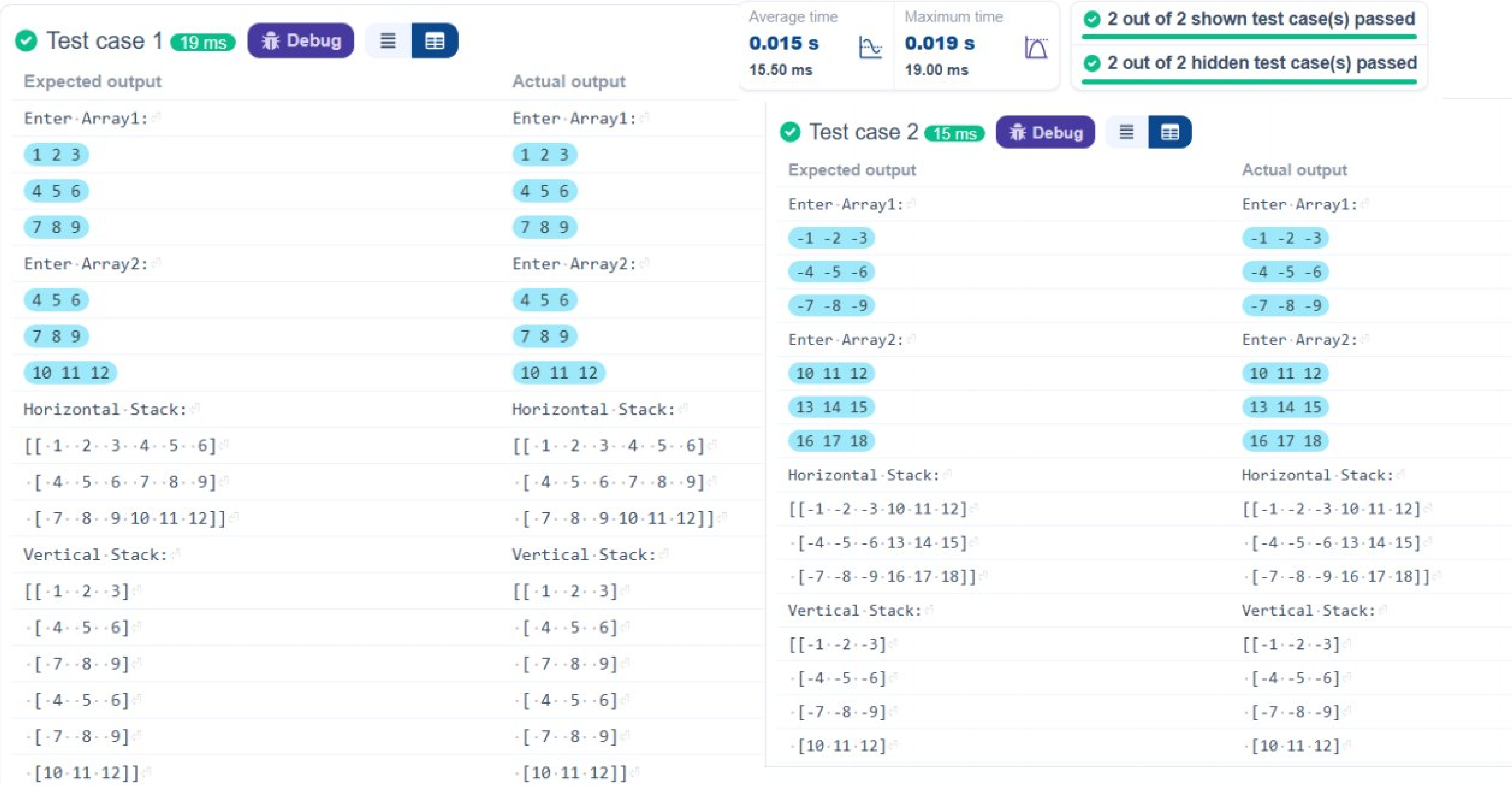
Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code:

```
import numpy as np

# Input matrices

print("Enter Array1:")

arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")

arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)

a=np.hstack((arr1,arr2))

print("Horizontal Stack:")

print(a)

# Perform vertical stacking (vstack)

b = np.vstack((arr1,arr2))

print("Vertical Stack:")

print(b)
```

Output:



3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Code:

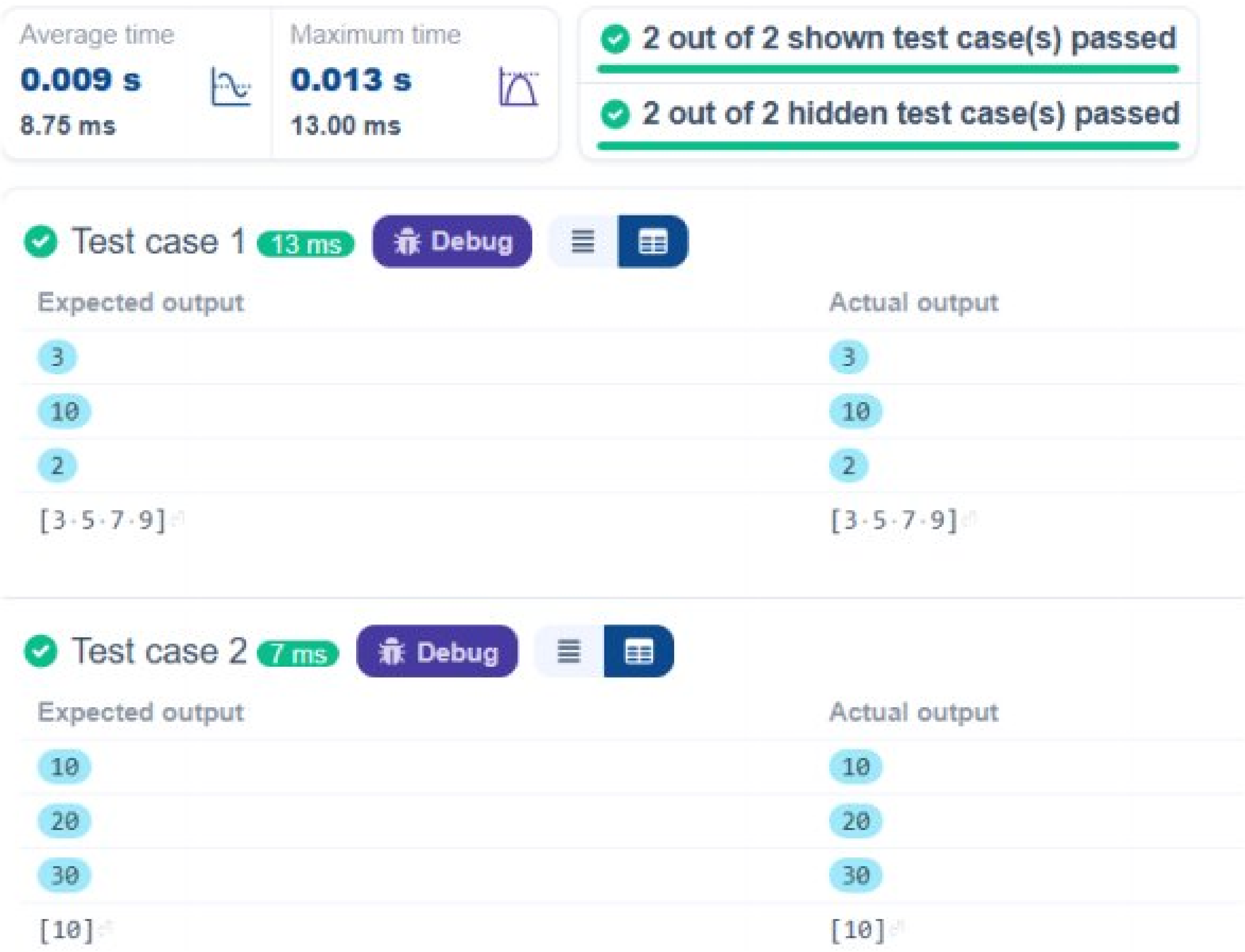
import numpy as np

 # Take user input for the start, stop, and step of the sequence

start = int(input())

```
stop = int(input())

step = int(input())

# Generate the sequence using np.arange()

a = np.arange(start,stop, step)

print(a)
```

Output:



3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

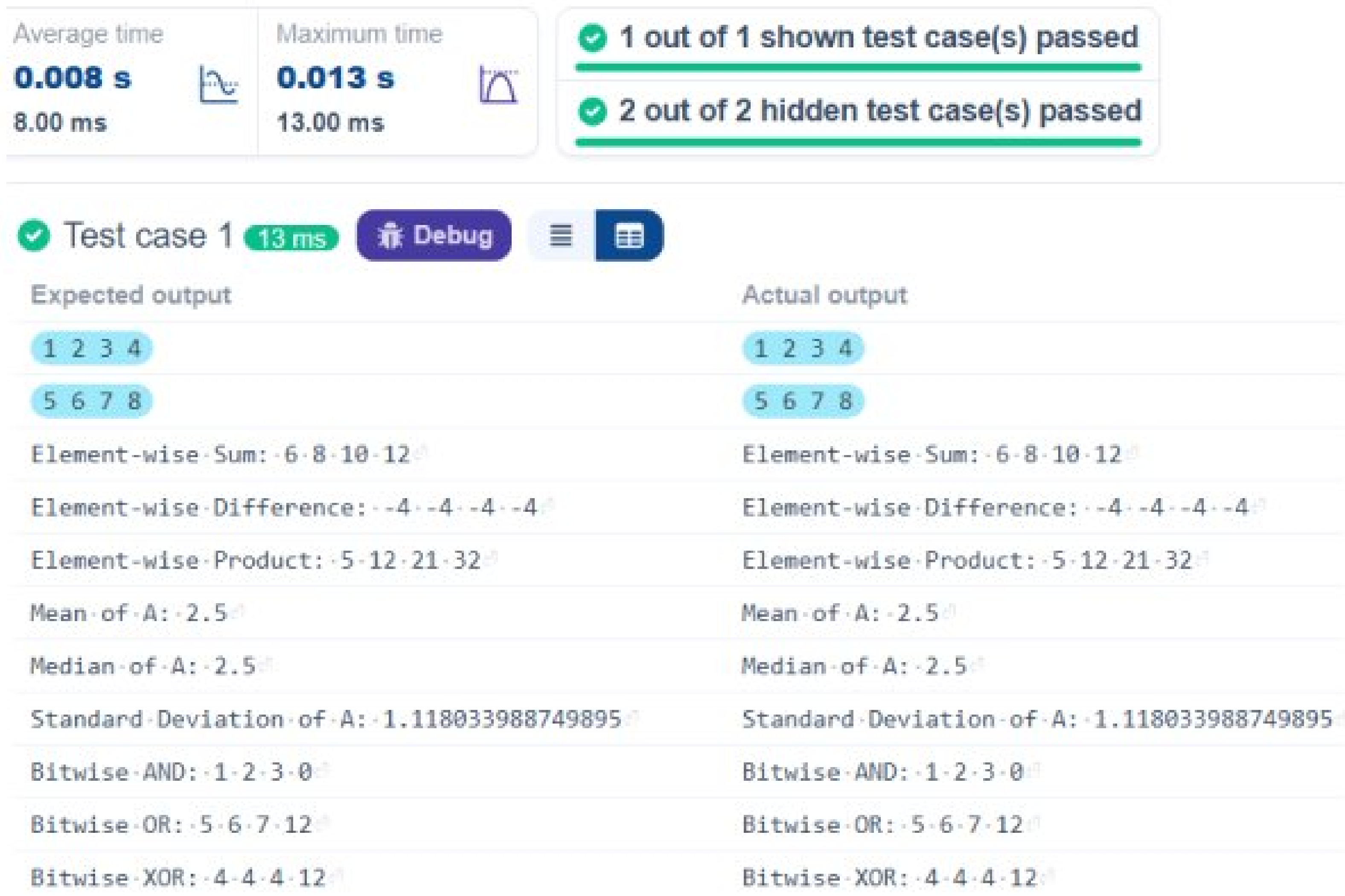
- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Code:

```
import numpy as np
def array_operations(A, B):

        # Convert A and B to NumPy arrays
        A = np.array(A)
        B = np.array(B)

        # Arithmetic Operations
        sum_result = A+B
        diff_result = A-B
        prod_result = A*B

        # Statistical Operations
        mean_A = np.mean(A)
        median_A = np.median(A)
        std_dev_A = np.std(A)

        # Bitwise Operations
        and_result = A & B
        or_result = A | B
        xor_result = A ^ B

  # Output results with one space between each element
        print("Element-wise Sum:", ' '.join(map(str, sum_result)))
        print("Element-wise Difference:", ' '.join(map(str, diff_result)))
        print("Element-wise Product:", ' '.join(map(str, prod_result)))

        print(f"Mean of A: {mean_A}")
        print(f"Median of A: {median_A}")
        print(f"Standard Deviation of A: {std_dev_A}")
```

```
    print("Bitwise AND:", ' '.join(map(str, and_result)))
    print("Bitwise OR:", ' '.join(map(str, or_result)))
    print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split()))  # Elements of array A
B = list(map(int, input().split()))  # Elements of array B
array_operations(A, B)
```

Output:



3.2.5. Numpy: Copying and Viewing Arrays

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <**original_array**>

View array: <**view_array**>

- After modifying the copy:

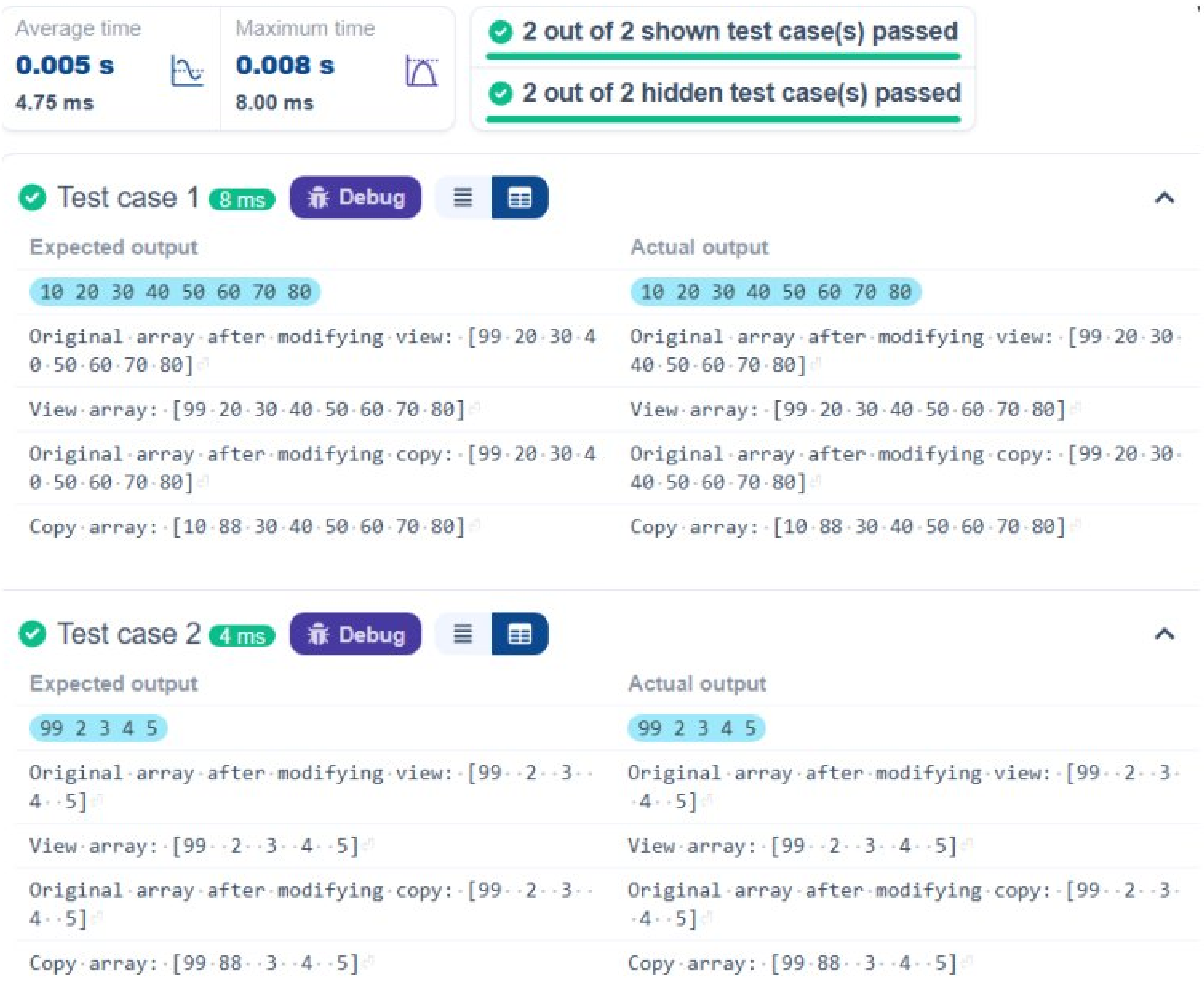
Original array after modifying copy: <**original_array**>

Copy array: <**copy_array**>

Code:

```
import numpy as np

inputlist = list(map(int,input().split(" ")))

# Original array

original_array = np.array(inputlist)

# Create a view

view_array = original_array.view()

# Create a copy

copy_array = original_array.copy()

# Modify the view

view_array[0] = 99

print("Original array after modifying view:", original_array)

print("View array:", view_array)

# Modify the copy

copy_array[1] = 88

print("Original array after modifying copy:", original_array)

print("Copy array:", copy_array)
```

Output:



3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. Searching: Find the indices where search_value appears in array1 and print these indices.

2. Counting: Count how many times count_value appears in array1 and print the count.

3. Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.

4. Sorting: Sort array1 in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.

3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

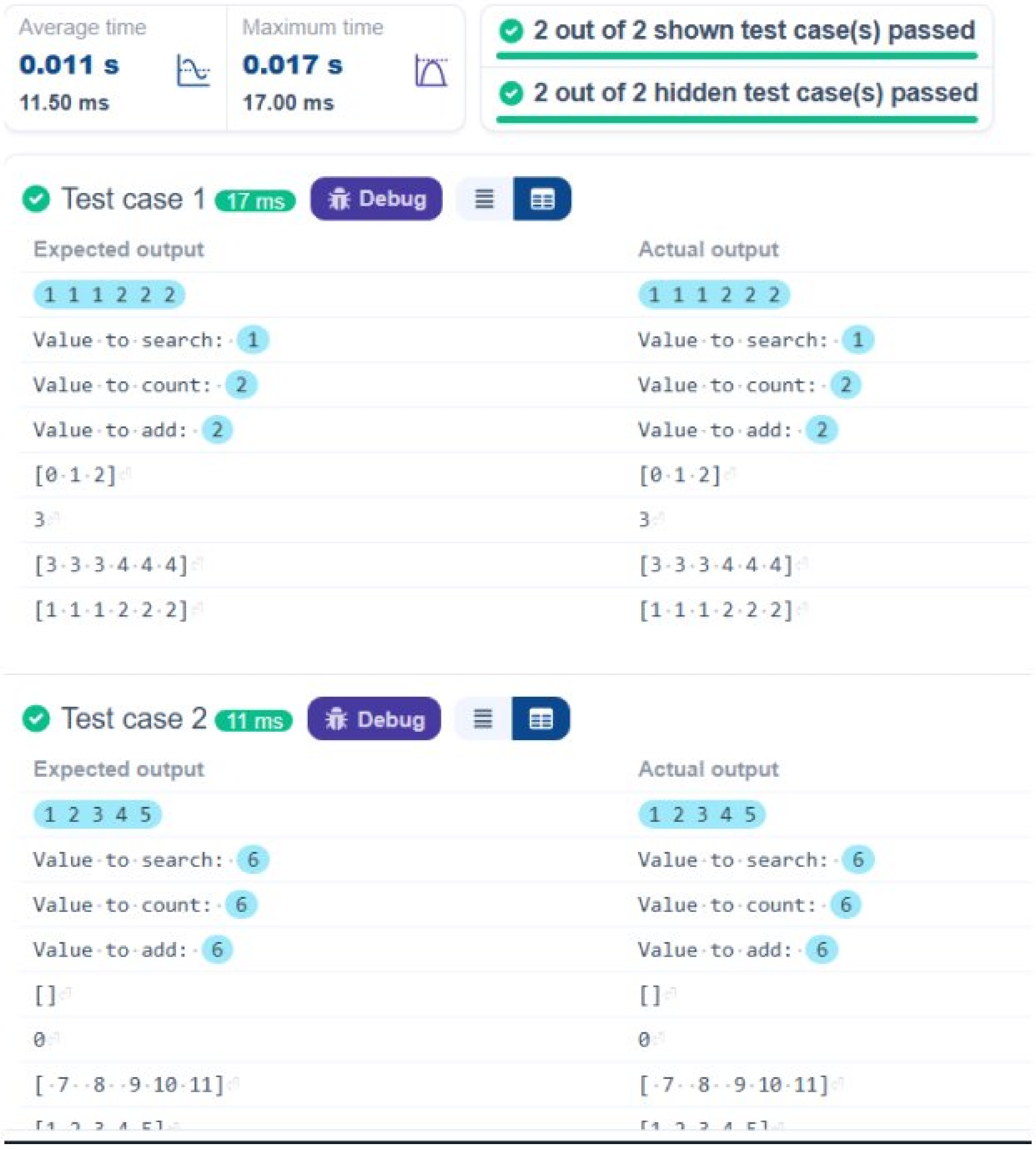
Output Format:

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

Code:

```
import numpy as np

# Input array from the user

array1 = np.array(list(map(int, input().split())))

# Searching

search_value = int(input("Value to search: "))

count_value = int(input("Value to count: "))

broadcast_value = int(input("Value to add: "))

# Find indices where value matches in array1

print(np.where(array1 == search_value)[0])

# Count occurrences in array1

print(np.count_nonzero(array1==count_value))

# Broadcasting addition

print(array1 + broadcast_value)

# Sort the first array

print(np.sort(array1))
```

Output:



3.2.7. Student Data Analysis and Operations

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.

- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1**: Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1**: Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

Code:

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# 1. Print all student details

print("All student Details:\n", a  )

# 2. print total students

print("Total Students:",len(a) )

# 3. Print all student Roll numbers

print("All Student Roll Nos", a[:,0]   )
```

```
# 4. Print subject 1 marks
print("Subject 1 Marks", a[:,1])
# 5. print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(a[:, 2])   )
# 6. print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(a[:,3])  )
# 7. Print All subject marks
print("All subject marks:",  a[:,1:] )
# 8. print Total marks of students
print("Total Marks", np.sum(a[:,1:], axis = 1)  )
# 9. print average marks of each student
print(np.round(np.mean(a[:,1:], axis = 1),1))
# 10. print average marks of each subject
print("Average Marks of each subject",  np.mean(a[:, 1:], axis = 0) )
# 11. print average marks of S1 and S2
print("Average Marks of S1 and S2", np.mean(a[:, 1:3],axis=0 )  )
# 12. print average marks of S1 and S3
print("Average Marks of S1 and S3",  np.mean(a[:, [1, 3]], axis = 0)  )
# 13. print Roll number who got maximum marks in Subject 3
max_sub3_index = np.argmax(a[:, 3])
print("Roll no who got maximum marks in Subject 3", a[max_sub3_index , 0] )
# 14. print Roll number who got minimum marks in Subject 2
min_sub2_index = np.argmin(a[:, 2])
print("Roll no who got minimum marks in Subject 2", a[min_sub2_index, 0] )
# 15. print Roll number who got 24 marks in Subject 2
print(f"Roll no who got 24 marks in Subject 2 [{a[a[:,2] == 24][:,0]}]" )
# 16. print count of students who got marks in Subject 1 < 40
count_sub1_lt_40 = np.sum(a[:,1]<40)
print("Count of students who got marks in Subject 1 < 40",  count_sub1_lt_40  )
# 17. print count of students who got marks in Subject 2 > 90
count__sub2_gt_90 = np.sum(a[:,2]>90)
```

```
print("Count of students who got marks in Subject 2 > 90:",  count__sub2_gt_90 )
# 18. print count of students in each subject who got marks >= 90
count_each_subject_90plus = np.sum(a[:,1:]>=90 , axis = 0)
print("Count of students in each subject who got marks >= 90:",   count_each_subject_90plus  )
# 19. print count of subjects in which each student got marks >= 90
print("Roll no:", a[:,0].astype(float)  )
print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:] >= 90 , axis = 1))
# 20. Print S1 marks in ascending order
print(np.sort(a[:,1]))
# 21. Print S1 marks >= 50 and <= 90
s1_filteredv = a[(a[:,1]>=50) & (a[:,1]<=90)]
print(s1_filteredv)
print(a)
# 22. Print the index position of marks 79
indices_79 = np.where(a[:,1:]==79)[0]
print((indices_79,))
```

Output:

Average time 0.017 s 17.00 ms | Maximum time 0.017 s 17.00 ms

1 out of 1 shown test case(s) passed

Test case 1 17 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
Total Students: 10	Total Students: 10
All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]	Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]
Min marks in Subject 2 12.0	Min marks in Subject 2 12.0
Max marks in Subject 3 99.0	Max marks in Subject 3 99.0
All subject marks: [[67. 77. 88.]	All subject marks: [[67. 77. 88.]
[78. 88. 77.]	[78. 88. 77.]
[89. 32. 88.]	[89. 32. 88.]
[79. 24. 33.]]	[79. 24. 33.]]
Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]	Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]
[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]	Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]	Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]	Average Marks of S1 and S3 [71.7 67.7]
Roll no who got maximum marks in Subject 3 305.0	Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 2 307.0	Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]	Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0	Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1	Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0 1 1]	Count of students in each subject who got marks >= 90: [0 1 1]
Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]	Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]	[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]